Quiz - Lecture 05

Close your laptops and put away all electronics. No electronics or notes are allowed during the duration of this quiz.

Question 1

What does the following code do?

```python
jupyturtle.make_turtle()
```

A. Imports the turtle module
B. Creates a window for drawing
C. Creates a turtle object
D. Moves the turtle forward

Question 2

Which statement best describes what `jupyturtle.left(90)` does?

A. Moves the turtle 90 units left
B. Rotates the turtle 90 degrees counterclockwise
C. Rotates the turtle 90 degrees clockwise
D. Turns the turtle to face west

Question 3

Why is this function definition useful?

```python
def square():
    for i in range(4):
        forward(100)
        left(90)
```

A. It reduces indentation errors
B. It avoids using loops
C. It makes the turtle move faster
D. It allows code reuse

Question 4

What shape does this code draw?

```python
for i in range(4):
    forward(50)
    left(90)
```

A. Triangle
B. Rectangle
C. Square
D. Circle

Question 5

If the turtle starts facing right (East), what direction is it facing after running this code?

```python
left(90)
left(90)
```

A. Up
B. Left
C. Down
D. Right

Question 6

How many times does `forward(100)` execute?

```python
for i in range(1):
    forward(100)
```

A. 0
B. 1
C. 100
D. Infinite

Question 7

What effect does changing `length` have?

```python
def line(length):
    forward(length)

line(200)
```

A. Changes the turtle's speed
B. Changes the turning angle
C. Changes how far the turtle moves
D. Changes pen thickness

Question 8

If the turtle starts by facing up (North), what direction is it facing after calling `turn()` once?

```python
def turn():
    right(90)
```

A. Up
B. Down
C. Left
D. Right

Question 9

What is a function?

A. A loop that repeats code
B. A variable that stores numbers
C. A drawing made by the turtle
D. A block of code that performs a task

Question 10

Why do we use functions?

A. To reuse code and keep programs organized
B. To make the turtle move faster
C. To avoid writing Python code
D. To remove loops

Bonus Question 1

What shape will this draw?

```python
def square():
    for i in range(3):
        forward(100)
        left(90)
```

A. Pentagon
B. Triangle
C. Square
D. Incomplete Square

Bonus Question 2

What shape will this draw?

```python
def square(length):
    for i in range(4):
        forward(length)
        left(90)

square(50)
square(100)
```

A. One square with side length 150
B. Two squares of the same size
C. One rectangle
D. Two squares of different sizes

Bonus Question 3

Write code that creates a turtle, then uses a for loop to draw 36 squares at size 100, rotating the turtle 10 degrees left after each square. You do not have to define the `square()` function. Assume `jupyturtle` is already imported.

```
jupyturtle.make_turtle()
for i in range(36):
    square()
    left(10)
```